
UNIT 14 SEARCH THEORY AND UNEMPLOYMENT*

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14.0 OBJECTIVES

After going through this unit you should be in a position to:

- explain the basic difference between Walrasian and non-Walrasian theories of unemployment;
- appreciate that the search and matching models are an extension of the neoclassical/ monetarist theories of employment and unemployment;
- appreciate the context in which the search theory of unemployment was developed;
- explain the use of a specific search and matching model to determine unemployment and its variation over time; and
- describe certain alternative search and matching models.

14.0 INTRODUCTION

A major problem with the labour market is that the demand for labour does not match the supply of labour, with an implication that certain percentage of the labour force remains unemployed. It is interesting to note that in the labour market there are workers without jobs while there are jobs without workers. Thus there are some unemployed workers who are in search of jobs and there are certain vacancies which are waiting to be filled up. Economists attribute it to

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various frictions and structural issues in an economy. The problem comes up because (i) labour as a category is heterogeneous, and (ii) certain costs (including money and time) are involved in the recruitment process. These characteristics of the labour market has prompted economists to propagate search and matching models of unemployment. The neoclassical theories have the drawback that they are unable to explain prolonged periods of involuntary unemployment experienced by an individual that characterise the real world. In classical theory the prevailing unemployment is always voluntary unemployment. Keynesian theories, on the other hand, explain the real world phenomenon of unemployment as involuntary unemployment by invoking the idea of deficiency of aggregate demand. Keynesian models, however, lack the elegance of the neoclassical models of employment and output. In particular, the Keynesian models do not provide microeconomic foundations for the rigidities in prices and wages that are postulated to show the existence of involuntary unemployment in the economy.

In this Unit we will examine further developments of these traditional theories. The main difference between the developments based on the Neoclassical theories and the developments springing from the Keynesian theories is that the former are based on the Walrasian general equilibrium model of perfectly competitive markets, whereas the latter are based on some non-Walrasian features introduced into the analysis, e.g., imperfect competition. In a Walrasian general equilibrium model, labour market is, like all other markets in the economy is a perfectly competitive one and there is no reason why the market should not clear. Involuntary unemployment is inconsistent with a Walrasian setting, so that if unemployment exists, it must be due to some non-Walrasian characteristics of the labour market.

The search theory that we are going to study in this unit becomes important because it shows that unemployment can exist as an equilibrium phenomenon even in a Walrasian setting of perfectly competitive labour markets. We begin, in Section 14.2, placing the search theory in the context of other theories of unemployment that you have studied, or are going to study in this block.

14.2 SEARCH THEORY AND THEORIES OF UNEMPLOYMENT

You need to understand the search and matching theories of unemployment in the context of other theories of unemployment. With this objective in view, we classify, in this section, the theories of unemployment that we are studying into four kinds.

If there is unemployment in a Walrasian labour market, unemployed workers would immediately bid down wages until supply and demand for labour are again in balance. If this process of bidding down wages is not working freely, there must be distinct reasons for it. We classify the theories of unemployment according to whether the process of wage adjustment is working or not, and

according to the reason why it is not working in cases where it does not work. In particular, consider an unemployed worker, who claims to be identical to a firm's current workers, and who offers to work for the firm at a marginally lower wage than the one the firm is currently paying its workers. There are four possible responses of the firm, giving us four kinds of theories of unemployment. These responses are:

- (i) If the firm accepts the worker's offer, we can conclude that the market for labour is Walrasian. In this view all observed unemployment is voluntary unemployment – unemployment of people moving between jobs and of those who are ready to work only at wages higher than the prevalent wage rate. This is really the *Neoclassical model* of unemployment referred to above.
- (ii) Secondly, the firm can respond to the unemployed worker's offer by saying that it does not accept the premise that the unemployed worker is identical to the firm's current employees. In this view, the labour market is not a market for a homogenous commodity, but is characterised by heterogeneity. Each job is unique and requires the unique skills that are embodied in an individual. The unemployed workers are matched with existing vacancies not through the market, but through a complex process of search and match. The models of unemployment that postulate such a process are called *search and matching models*.
- (iii) Thirdly, the firm can respond by saying that, even though it would like to cut the wages and employ the additional worker, it cannot do this because it is bound by implicit and explicit agreements with its workers, arrived at through collective bargaining, regarding the wages that have to be paid. Wages are thus institutionally determined in these models known as *contracting models*.
- (iv) Lastly, the firm may respond by saying that it does not want to reduce real wages – it believes that the benefits accruing to it from higher wages are more than the costs of maintaining wages high. Theories that build up on this idea are called *efficiency-wage theories* in an obvious reference to the fact that higher wages impart benefits to the employing firm by improving the efficiency of labour.

In this Unit, you will study in details one of the search and matching models of unemployment referred to in (ii) above. You will study the contracting models and the efficiency-wage theories of unemployment, referred to respectively in (iii) and (iv) above, under the rubric of the New Keynesian Theories of Unemployment in Unit 16.

Check Your Progress 1

1. Why is a Walrasian general equilibrium model inconsistent with unemployment?

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2. Suggest a classification of theories of unemployment based on the postulated responses of a firm to an offer by an unemployed worker to work for it at a slightly lower wage.

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14.3 SEARCH THEORIES – A BRIEF HISTORICAL OVERVIEW

A search theory of unemployment is found even in the writings of A.C. Pigou in the inter-war period. To explain the high unemployment in the period Pigou used an idea you are very familiar with – that workers are unemployed because wages are too high. Keynes contested this idea in the development of his *General Theory of Employment, Interest and Money*. But initially Pigou had tried to explain the high unemployment of the inter-war period with reference to another idea – the idea of frictional unemployment, where unemployment arises as workers shift between jobs, moving to jobs where their productivity is higher. Search and matching unemployment is actually a form of frictional unemployment – unemployment which arises because of the frictions in shifting between jobs generated by the fact that skills are to be matched with vacancies in the job ‘market’. Pigou himself was aware though that jobs were not shifting around too much in the 1920s, so that he ultimately banked more on ‘workers pricing themselves out of the market through trade union activities’ as an explanation for the inter-war unemployment.

The idea of search unemployment was subsequently formalised in the 1970s and 1980s to make the Neo-classical Walrasian model accord with the reality of the empirically observed and varying unemployment in the labour market, as has

been indicated to you in the introductory section. The importance of search in decentralised markets was first emphasized in an influential book edited by Edmund Phelps in 1970 (*Microeconomic Foundations of Employment and Inflation Theory*, Norton, New York). This book contains some of the first formal models using search theory to explain unemployment as an equilibrium phenomenon. Lucas and Prescott presented in 1974 a general equilibrium model of unemployment. In this model stochastic sectoral shocks induce workers to move between sectors, but there is a one-period lag by workers in moving between sectors, brought about through search and matching kind of considerations. Unemployment is generated in the model by this lag.

In the 1980s, search models were built up as continuous time general equilibrium models, in the tradition of the models built under the real business cycle theory. Noteworthy amongst these are the models by Peter Diamond: the paper titled “Mobility Costs, Frictional Unemployment, and Efficiency” published in the *Journal of Political Economy* in 1981, and by Christopher Pissarides: the paper titled “Short-Run Equilibrium Dynamics of Unemployment, Vacancies, and Real Wages” published in the *American Economic Review* in 1985. We have reproduced the titles of the papers for you because they provide a flavour of the concepts and mechanisms used to rationalize unemployment as an equilibrium phenomenon in a Walrasian model. We will examine, very briefly, the model by Pissarides in Section 14.5. Before that, however, let us develop a more complete model of search unemployment in the next section.

Check Your Progress 2

1. Why was it necessary to augment the neoclassical/ monetarist theory of employment through search and matching kind of models?

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2. What is frictional unemployment?

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14.3 A SEARCH AND MATCHING MODEL

It should be clear to you from the earlier section that there are a variety of models under the rubric of search theory. In this section we examine one such model at close quarters. Peter Howitt originally developed the model as “Business Cycles with Costly Search and Recruiting” in the *Quarterly Journal of Econometrics* in 1988. The exposition here is based on Blanchard and Fischer (2000). Unlike in other sections of this unit, the exposition in this section is necessarily more technical. It is important to follow it through, perhaps with the help of the book, in order to get a flavour of the kind of analysis that you will find in the literature today. The approach is descriptive and the use of equations is minimised. Going through equations, however, can add to the understanding of the expounded ideas and you are advised to follow the equations-based exposition of the model in Blanchard and Fischer (2000).

14.4.1 Model Specification

We proceed with the model specification in the following steps.

1. The economy is composed of competitive firms (F in number) and identical workers (N in number). In each discrete time period a fraction δ of the employed is laid off and joins the unemployment pool. The fraction δ is called the ‘rate of separation’ in the literature. Firms hire workers from the pool, not directly from other firms.
2. The marginal cost of hiring for each firm is an increasing function of its level of hiring. This captures the idea that a high rate of hiring may force firms to increase their search intensity or, in a more general model with heterogeneous workers, to accept poor matches between workers and jobs. The marginal cost is also a decreasing function of aggregate unemployment – high aggregate unemployment makes it easier and cheaper for the firm to find willing and competent workers.
3. Since each firm chooses the rate of hiring by equating the marginal cost of hiring to the net marginal benefit of hiring, it is important to determine, in the model, the marginal benefit of hiring to the firm. Assuming a firm to be risk neutral, the marginal value to the firm of a worker hired in this period is the expected present value of his marginal product so long as he works with the firm. The marginal value, denoted by q_t , is therefore an infinite sum of discounted marginal productivities from the present period onwards to infinity. Two discounting factors are used on each term: one, as usual, to take account of time and the other to take account of the probability that a given worker will have left the job by time $(t + i)$.
4. The net marginal benefit of hiring is equal to this marginal value *minus* the discounted present value of wages to be paid to the worker who is newly hired. It is in the spirit of search and matching models to assume that there is no labour market in which the wage is set – job matches require an explicit search process.

The wage is set through bargaining so as to divide the surplus from the job between the worker and the firm. To simplify matters, it is assumed in the present model that the worker experiences neither costs nor benefits from unemployment, so that the total surplus from the job is just the marginal value determined in paragraph 3 above. It is assumed that the worker obtains a share ξ of the surplus and the firm gets $(1 - \xi)$ with the size of ξ reflecting the bargaining power of the worker. Thus the marginal benefit of hiring to the firm is given as a fraction of the marginal value q_t :

$$\lambda_t = (1 - \xi) \cdot q_t$$

Each firm chooses the rate of hiring, h_t , by equating the marginal benefit of hiring specified in paragraph 4 above with the marginal cost of hiring determined in paragraph 2 above

14.4.2 Model Solution and the Equilibrium Rate of Unemployment

Given the above specification, the model can be solved for the marginal benefit of hiring, λ_t , and the hiring rate, h_t . If the employment in the firm is denoted by n_t , it follows that $n_t = (1 - \delta) \cdot n_{t-1} + h_t$, since employment in period t is given by employment in period $(t - 1)$, as adjusted for the rate of separation and the rate of hiring. Assuming that there are F identical firms in the economy, the unemployment rate, denoted by u_t , is given by $1 - (F \cdot n_t)/N$, where N is the total number of workers in the economy. These four equations for λ_t , h_t , n_t and u_t can be solved to obtain the equation characterising the dynamics of the equilibrium unemployment rate:

$$u_t = \delta + \{1 - \delta - (F/G)\lambda_t\} \cdot u_{t-1}$$

In the above equation, G is a parameter in the cost of hiring function, such that a larger parameter value denotes higher difficulty of locating workers. The equation clearly shows that the unemployment rate depends on its own lagged value and the constant rate of separation, δ . It also depends on the state of technology via its dependence on the net marginal benefit of hiring, λ_t , since the latter depends on the marginal product of worker.

When the marginal productivity of labour is postulated to have zero variance, the natural rate of unemployment is given by

$$u^* = \delta / \{\delta + (F/G) \cdot k\}$$

A clear result emerges from this equation: the larger the separation rate, δ , and the larger the parameter G (reflecting the difficulty of locating workers), the higher is the rate of unemployment.

14.4.3 Optimality of the Equilibrium Unemployment Rate

The above model rigorously rationalizes the existence of unemployment. As we will see below, shocks to productivity can also be used to explain the variability of the equilibrium rate of unemployment over time. You should, however

appreciate that the equilibrium rate of unemployment obtained in the above model is unlikely to be socially optimal. This is so for two reasons:

- (1) Hiring decision by a firm is beneficial to it to the extent that the net marginal benefit of hiring is positive, but it imposes a cost on other firms, an externality that is not taken into account by the hiring firm. By hiring an extra worker, the firm decreases unemployment, and since the marginal cost of hiring is a decreasing function of aggregate unemployment, the marginal cost of hiring to other firms is increased. This effect leads to too much hiring compared to the social optimum, and thus to too low an equilibrium rate of unemployment.
- (2) There is also a divergence between the social and private marginal benefit of hiring: the former is given by q_t , whereas the latter (the private benefit to the hiring firm) is given by a fraction $(1 - \xi)$ of q_t , depending on the bargaining power of the worker *vis-à-vis* the hiring firm. Since the private benefit is less than the social benefit ($\xi > 0$), there is too little hiring and thus too high an equilibrium rate of unemployment.

The two effects, in (1) and (2) above, work in opposite directions, one tending to increase the equilibrium rate above the socially optimum rate and the other tending to keep it below the social optimum. The net effect on the equilibrium rate of unemployment *vis-à-vis* the socially optimum rate is ambiguous in the model.

14.4.4 Dynamics of Unemployment and Real Wages through Productivity Shocks

The model that you are studying here is in the tradition of the real business cycle theory that you have studied in earlier units. As you know, this kind of a model works out the implications of shocks to productivity. The model has the following implications to employment and wages.

- (1) A temporary adverse shock to productivity decreases hiring (as it decreases the marginal productivity of labour and hence the benefit of hiring the marginal unit of labour) and increases unemployment. As the shock is, by definition, temporary, productivity and the net marginal value of labour return to their original level, but, it can be shown that the unemployment rate only slowly returns to normal through increased hiring. Moreover, since it is cheaper for the firm to hire when there are more unemployed, a productivity shock has greater effect on unemployment when it is high than when it is low. This is, of course, implicit in the non-linearity of the equation explaining u^* , the natural rate of unemployment.

- (2) The model explains why fluctuations in employment may be associated with smaller fluctuations in real wages. This will happen if ξ , the share obtained by workers, is constant, as is assumed in the model, and small. Real wages vary in the model with productivity and high rates of hiring are associated with high real wages. The model thus explains the observed empirical fact of a pro-cyclical increase in real wages, but to a smaller extent than the increase in employment, if the share obtained by workers is small in relation to that obtained by the hiring firms.

Check Your Progress 3

1. Explain why introduction of search and matching introduces equilibrium unemployment in a Walrasian model.

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2. Why is the equilibrium unemployment rate, obtained in the Howitt (1988) model, unlikely to be socially optimal?

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3. How does the Howitt (1988) model explain

- a. An increase in unemployment
- b. A smaller pro-cyclical response of real wages in relation to employment?

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14.5 SOME ALTERNATIVE SEARCH MODELS

In this section we look, briefly, at two more papers emphasizing frictions arising due to search and matching considerations: one by Pissarides that has been already referred to and another by David Lilien. The exposition here is again based on Blanchard and Fischer (2000).

The model developed in Section 14.4 is just one of the many search and matching models developed in the literature. In this section we examine, briefly, some alternative search and matching models. One of the shortcomings of the Howitt (1988) model that was discussed in Section 14.4 is that it simply postulates that search and matching is undertaken by workers and firms, but does not specify the search technology and the matching process used by workers and firms. Also the Howitt model takes the share of workers parameter, ξ , as given. Pissarides developed a closely related model in 1985 wherein these shortcomings were addressed. The model clearly shows that unemployment emerges as an equilibrium phenomenon in an otherwise neoclassical model that is characterised by workers moving from one job to another and remaining unemployed in the interim during their search for the right kind of job as a replacement for the job they have discarded. The approach shows explicitly the dependence of the rate of hiring on the characteristics of the labour markets. Moreover, unlike in the Howitt model, the Pissarides model helps in thinking about what determines the share of workers parameter ξ in the bargaining process between the workers and the firms – the parameter value depends on the option that the workers have in turning down the match and looking for another match.

Models have also been developed to capture the effects of sectoral shocks – changes in relative productivity or changes in relative demand for goods – on aggregate equilibrium unemployment. David Lilien emphasized frictions arising due to the inability of labour to relocate instantaneously and costlessly between sectors in a paper titled “Sectoral Shifts and Cyclical Unemployment” published in the *Journal of Political Economy* in 1982. Consider an economy with two sectors. Labour is immobile between sectors within periods but fully mobile across sectors over periods. Workers in each sector supply one unit of labour inelastically if the wage exceeds a reservation wage. Assume that the wage is sufficiently higher than the reservation wage in both sectors and that labour is fully employed between the two sectors. Assume further that labour demand shifts away from sector 1 toward sector 2, so that within the period wage increases in sector 2 and falls in sector 1. Since labour cannot shift from sector 1 to sector 2 within the period, employment cannot increase in sector 2, but falls in sector 1 due to the decrease in labour demand. The sectoral shift hence increases unemployment in the current period. In the following period labour reallocates itself and aggregate employment returns to normal.